COMP721: MACHINE LEARNING (2026)

SEMESTER PROJECT (150 MARKS)

Due Dates:
Phase 1 (Proposal): 14 August 2026; Time: 23:59
Phase 2 Submission (50% Completion): 11 September 2026; Time: 23:59
Phase 3 Submission (Final): October 17th, 2026; Time: 23:59
Final Presentation: October 26th, 2026; Time: 9:00 am – 12:00; Venue: Honors Lab

Note: Please consult the Submission Guide and Assessment Milestones document for comprehensive details on the project submission requirements, deliverables, deadlines, and assessment procedures.

Project Title: Cardiovascular Disease Risk Prediction Using Classical Machine Learning, Ensemble Learning, and Explainable Artificial Intelligence

1. Background

Cardiovascular disease (CVD) remains one of the leading causes of death worldwide. Early prediction of cardiovascular disease enables healthcare professionals to identify high-risk patients and implement preventive interventions. Machine learning has become an important tool for developing predictive healthcare systems. However, successful machine learning involves much more than simply training a classifier. It requires careful data preprocessing, feature engineering, feature selection, hyperparameter optimization, model evaluation, explainability, and statistical validation. In this project, your group will develop a complete end-to-end machine learning pipeline for cardiovascular disease prediction using classical machine learning algorithms.

2. Learning Outcomes

Upon successful completion of this project, you should be able to:

- Apply a complete machine learning workflow to a real-world problem.
- Perform data cleaning and preprocessing.
- Engineer meaningful features from existing variables.
- Apply feature selection techniques.
- Train classical machine learning models from scratch.
- Perform hyperparameter optimization using cross-validation.
- Develop an ensemble learning model.
- Evaluate models using appropriate performance metrics.
- Explain model predictions using SHAP.
- Perform statistical analysis to compare model performance.
- Communicate findings through a scientific report.

3. Project Description

Develop a machine learning system capable of predicting whether a patient is at risk of cardiovascular disease using demographic, physiological, and lifestyle information. You are required to investigate whether an ensemble learning model provides statistically significant improvements over individual machine learning models.

4. Dataset

Students must use the following dataset.

Cardiovascular Disease Dataset

https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset

Dataset Characteristics

- Approximately 70,000 patient records
- 13 variables
- Binary classification problem
- Publicly available
- CSV format

5. Group Requirements

- Students must work in groups of **four (4)**.
- Every member must contribute equally.
- All members must understand every aspect of the project.
- Every member will participate in the final presentation.

6. Software Requirements

Students may use

- Python
- Google Colab
- Jupyter Notebook
- Scikit-learn
- NumPy
- Pandas
- Matplotlib
- SHAP

No deep learning frameworks are required.

7. Project Requirements

Phase 1: Data Understanding

Students must:

- Describe the dataset.
- Explain every feature.
- Identify the target variable.
- Determine the class distribution.
- Identify data types.
- Produce summary statistics.

Required outputs

- Dataset description
- Feature description table
- Descriptive statistics
- Histograms
- Correlation heatmap
- Class distribution plot

Phase 2: Data Preprocessing

Students must perform appropriate preprocessing.

This should include:

- Duplicate removal
- Missing value analysis
- Missing value imputation (if applicable)
- Detection and removal/treatment of outliers
- Handling inconsistent values
- Encoding categorical variables
- Feature scaling where necessary

Students must justify every preprocessing step.

Phase 3: Exploratory Data Analysis (EDA)

Students should investigate relationships within the data.

Examples include:

- Correlation analysis
- Feature distributions

- Boxplots
- Scatter plots
- Pairwise relationships
- Class imbalance analysis

Students should discuss important observations.

Phase 4: Feature Engineering

Students must create **at least five (5)** new features.

At least **three (3)** of these must be medically meaningful and supported by literature.

Possible examples include:

- Body Mass Index (BMI)
- Pulse Pressure
- Mean Arterial Pressure
- Hypertension Stage
- BMI Category
- Lifestyle Risk Score
- Metabolic Risk Score
- Cardiovascular Health Index
- Age Category

Students must explain:

- how each feature was created,
- why it is clinically meaningful, and
- why it may improve prediction performance.

Phase 5: Feature Selection

Students must apply **at least one** feature selection technique.

Acceptable methods include:

- Recursive Feature Elimination (RFE)
- Mutual Information
- ANOVA F-score
- SelectKBest

Students should compare:

- original features,
- engineered features,

- selected features.

Students must justify why the final feature subset was chosen.

8. Machine Learning Models

Students must develop the following models.

Model 1: Decision Tree Classifier

Model 2: Support Vector Machine (SVM)

Model 3:

Ensemble Model

Students may choose ONE of the following:

- Soft Voting Classifier
- Stacking Classifier

The ensemble must combine Models 1 and 2.

9. Hyperparameter Optimization

Each individual model must be trained using **four (4)** different hyperparameter configurations.

Example (Decision Tree)

- Maximum Depth = 5
- Maximum Depth = 10
- Maximum Depth = 15
- Maximum Depth = 20

Example (SVM)

- Linear Kernel
- Polynomial Kernel
- RBF Kernel
- Sigmoid Kernel

Students may define additional hyperparameters where appropriate but must clearly justify their choices.

10. Model Development Protocol

Students **must** follow the workflow below.

Step 1

Split the dataset into:

- Training Set (80%)
- Test Set (20%)

The test set must remain unseen until the final evaluation.

Step 2: Using only the training data, perform **5-fold cross-validation**.

Step 3: Evaluate all four hyperparameter configurations.

Step 4: Select the best hyperparameters using the average cross-validation performance.

Step 5: Retrain the model using the **entire training set** and the selected hyperparameters.

Step 6

Evaluate the final model **once** on the unseen test set.

No further tuning may be performed after evaluating the test set.

11. Model Evaluation

Students must report the following metrics.

- Classification Accuracy
- Precision
- Recall
- Specificity
- F1-score
- ROC-AUC
- Matthews Correlation Coefficient (MCC)
- Cohen's Kappa (optional but encouraged)

Required plots include:

- Confusion Matrix
- ROC Curve
- Precision–Recall Curve
- Feature Importance (Decision Tree)
- SHAP Summary Plot
- SHAP Bar Plot
- SHAP Dependence Plot
- Learning Curve (optional but encouraged)
- Validation performance for the four hyperparameter configurations

12. Explainable AI (SHAP)

Students must explain model predictions using SHAP.

The analysis must include:

- Global feature importance
- Local explanations for individual patients
- Comparison of important features identified by each model
- Discussion of clinically important variables

Students should interpret the results rather than simply presenting SHAP figures.

13. Statistical Analysis

Students must investigate whether the ensemble model performs significantly better than the individual models.

The analysis should follow these steps:

1. Use the performance scores obtained from the five cross-validation folds.
2. Assess the normality of the paired performance differences (e.g., using the Shapiro–Wilk test).
3. If the normality assumption is satisfied, perform a paired *t*-test.
4. Otherwise, perform the Wilcoxon signed-rank test.

The following comparisons are required:

- Decision Tree vs Ensemble
- SVM vs Ensemble

Students must report:

- null hypothesis,
- alternative hypothesis,
- significance level ($\alpha=0.05$\alpha = 0.05$\alpha=0.05$),
- test statistic,
- *p*-value,
- conclusion.

14. Discussion

Students should critically discuss:

- Which model performed best?
- Why did it perform best?
- Did feature engineering improve performance?
- Did feature selection improve performance?
- Which features were most influential?
- Did SHAP explanations agree with domain knowledge?
- Did the ensemble significantly outperform the individual models?
- What are the limitations of the study?
- What future improvements can be made?

15. Deliverables

Each group must submit:

- Source code (Python)
- Google Colab Notebook
- Final cleaned dataset (if modified)
- Final report (PDF)
- Presentation slides

16. Final Report

The report should follow the structure below.

1. Title Page
2. Abstract

3. Introduction
4. Literature Review
5. Methodology
6. Experimental Setup
7. Results
8. Discussion
9. Conclusion
10. References
11. Appendices (if applicable)

Maximum length:

Approximately **25–40 pages**, excluding appendices.

17. Presentation

Each group will deliver a **10-minute presentation**, followed by **2 minutes of questions**.

Every group member must participate in the presentation or Q&A.

18. Assessment Criteria (150 Marks)

Component	**Due Date (Time)**	**Marks**
Phase 1 – Project Proposal	August 14th (23:59)	10
Phase 2 – Progress Report & Partial Code	September 11th (23:59)	30
Phase 3 – Final Technical Report & Complete Submission	October 17th (23:59)	60
Final Presentation	October 26th (9:00 – 12:00)	50
Total		**150**

Academic Integrity

- All submitted work must be the original work of the group.
- All external sources, code libraries, datasets, and references must be appropriately cited.
- The use of generative AI tools (e.g., ChatGPT, Copilot) is permitted as a learning aid. Students remain fully responsible for the correctness, originality, implementation, analysis, and interpretation of all submitted work. Any AI assistance used must be acknowledged in an appendix describing how it was used.
- Plagiarism, fabricated results, copied code without attribution, or misrepresentation of contributions will be treated as academic misconduct in accordance with University regulations.